Yiliang Wan

Email: yiliangw@comp.nus.edu.sg
Mobile: 65-91757307

Education

National University of Singapore January 2024 – Present
Ph.D. in Computer Science (Advisor: Jialin Li) *Singapore*

- Research topics: Distributed Systems and Datacenter Networks

National University of Singapore August 2022 – December 2023
M.Sc. in Computer Science (Advisor: Jialin Li) *Singapore*

- GPA: 5.0/5.0
- Thesis: *Accelerating Consensus with Network Ordering in Wireless Networks*

Beihang University September 2017 – June 2022
B.Eng. in Computer Science and Technology *Beijing, China*

- GPA: 3.83/4.00

Ongoing Research

Efficient SMR with Practical Network Synchrony [Under Submission]

Practical distributed protocols assume weak network models, leaving significant performance unexploited. We show that datacenter networks can be engineered to provide strong synchrony with a tight round bound under 2μs, and co-design a replication protocol that exploits this synchrony to pipeline instances and allow parallel proposals in the common case, achieving up to 255% higher throughput over state-of-the-art protocols without compromising latency. [preprint][poster]

Full-Stack Simulation for CXL-Enabled Clusters [In Progress]

Evaluating datacenter software-hardware co-design requires end-to-end experimentation, yet existing simulators trade off between fidelity and speed. We design a simulation-oriented software stack centered on OS-level support, extending the Linux kernel and QEMU/KVM with simulation-aware scheduling, timing control, and performance modeling, enabling scalable end-to-end evaluation of unmodified datacenter software stacks with CXL-enabled disaggregated memory. [preprint]

Inference Systems for Generative Recommendation [In Progress]

Generative recommendation models pose new serving challenges distinct from LLMs, including sub-100ms latency requirements, massive per-user KV caches, and fragmented traffic patterns. We are working on a GPU-CPU co-serving framework to address these challenges.

Preprints

LiveStack: OS Support for Cluster-Scale Full-Stack Live Simulation
Yiliang Wan, Haifeng Sun, Yihan Yang, Jonas Kaufmann, Anotine Kaufmann, Jialin Li.
arXiv:2606.18958, arXiv 2026

Reaching Agreement Among Reasoning LLM Agents
Chaoyi Ruan, Yiliang Wan, Ziji Shi, Jialin Li.
arXiv:2512.20184, arXiv 2025

Building State Machine Replication Using Practical Network Synchrony
Yiliang Wan, Nitin Shivaraman, Akshaye Shenoi, Xiang Liu, Tao Luo, Jialin Li.
arXiv:2507.12792, arXiv 2025

Experience

Huawei - Research Intern July 2025 – February 2026
Markov Lab - Datacenter Simulation and Generative Recommendation Systems *Shanghai, China & Singapore*

Other Projects

Network-Ordered Consensus for Wireless Networks November 2022 – November 2023

National University of Singapore *Singapore*

- Co-designed consensus protocols with network ordering in wireless sensor networks.

High-Performance IPC for Microkernels October 2021 – June 2022

Beihang University *Beijing, China*

- Optimized message passing for AArch64 and achieved a 15% smaller latency than seL4.
- Introduced the IPC routine mechanism to reduce the overhead of user-space services.
- Implemented POSIX-compliant IPC services.

Embedded Hypervisor August 2021 – October 2022

Beihang University *Beijing, China*

- Developed device drivers for an embedded type-1 hypervisor on AArch64.

Teaching

Distributed Systems (CS5223)

National University of Singapore

Teaching Assistant Spring 2026

Teaching Assistant Spring 2025

Teaching Assistant Spring 2024

Introduction to Operating Systems (CS2106)

National University of Singapore

Teaching Assistant Fall 2024

Awards

NUS Research Scholarship 2024

National University of Singapore

Academic Excellence Award 2023

National University of Singapore

Outstanding Student Award 2021

Beihang University